

Frances V. Davenport

Colorado State University
email: f.davenport@colostate.edu
web: <https://fdavenport.github.io/>

April 2023 – present *Assistant Professor*, Dept of Civil and Environmental Engineering, Colorado State University
2022 – 2023 *Postdoctoral Fellow*, Dept of Atmospheric Science, Colorado State University
2017 – 2022 *Doctoral Student Researcher*, Dept of Earth System Science, Stanford University
2014 – 2017 *Civil Engineer I*, McLaughlin Water Engineers, Denver, CO

Education

Ph.D. **Stanford University**, Earth System Science, 3/2022
B.E. **Dartmouth College**, Thayer School of Engineering, Engineering, 3/2014
B.A. **Dartmouth College**, Engineering Sciences with High Honors, 6/2013

Peer-Reviewed Publications

Google Scholar [profile](#)

♦ denotes papers where Davenport is lead or senior author (i.e. last author).

Underlined denotes students or postdocs advised by Davenport (graduate student*, postdoc[±])

Submitted/In Review

- ♦ Keeney, N. J.* and **F. V. Davenport**, “A spatially and temporally disaggregated inland flood dataset with flood metrics (2000-2024)”, Under review at *Earth System Science Datasets*

Choi, E., **F. V. Davenport**, Z. Lin, A. Ortiz-Bobea, K. Reed, E. Kebeab, N. D. Mueller.
“Weather-driven US milk yield losses and economic damages revealed by 9 million cows”.
Under second review at *Nature Food*

Hasan, M. F., R. G. Smith, **F. V. Davenport**, & S. Majumdar. “Groundwater Pumping and Irrigation in the Western United States Estimated at High Resolution with Machine Learning and Satellite Products”. In revision for *Journal of Hydrology: Regional Studies*

Published

- ♦ 19. **Davenport, F. V.** and D. Touma (2026). Discrepancies in Extreme Daily and Monthly Precipitation Trends between Observations and Earth System Models. *Geophysical Research Letters*, 53, e2026GL124747. <https://doi.org/10.1029/2026GL124747>
- 18. Trok, J. T., E. A. Barnes, E. M. Gordon, **F. V. Davenport**, & N. S. Diffenbaugh (2026). Global emissions since the Paris agreement have intensified Europe's recent heatwaves. *Geophysical Research Letters*, 53, e2026GL125079. <https://doi.org/10.1029/2026GL125079>
- 17. Shaffer, M., N. Awad, **F. V. Davenport**, and S. Fakhreddine. (2026). Evaluating the Resilience of Drinking Water Systems to Severe Weather Events Using Boil Water Notices and Bottled Water Sales. *Environmental Science & Technology*, 60 (31), <https://doi.org/10.1021/acs.est.5c18579>

16. Orihuela-Pinto, B.[±], P. W. Keys, **F. V. Davenport**, and E. A. Barnes. (2025). Uncovering patterns of converging human-induced pressure on global lands. *Machine Learning: Earth*, 1, <https://doi.org/10.1088/3049-4753/ae2278>
15. Kalashnikov, D. A., J. T. Abatzoglou, **F. V. Davenport**, Z. M. Labe, P. C. Loikith, D. Touma, & D. Singh. (2025). Projections of lightning-ignited wildfire risk in the western United States. *Earth's Future*, 13, e2025EF006108. <https://doi.org/10.1029/2025EF006108>
14. Kalashnikov, D. A., **F. V. Davenport**, Z. M. Labe, P. C. Loikith, J. T. Abatzoglou, & D. Singh (2024). Predicting cloud-to-ground lightning in the western United States from the large-scale environment using explainable neural networks. *Journal of Geophysical Research: Atmospheres*, 129, e2024JD042147. <https://doi.org/10.1029/2024JD042147>
13. Trok, J. T., E. A. Barnes, **F. V. Davenport**, and N. S. Diffenbaugh (2024). Machine learning–based extreme event attribution, *Science Advances*, <https://doi.org/10.1126/sciadv.adl3242>.
- ♦ 12. **Davenport, F. V.**, E. A. Barnes, and E. M. Gordon (2024). Combining Neural Networks and CMIP6 Simulations to Learn Windows of Opportunity for Skillful Prediction of Multiyear Sea Surface Temperature Variability, *Geophysical Research Letters*, 51, e2023GL108099. <https://doi.org/10.1029/2023GL108099>
11. Gordon, E. M., E. A. Barnes and **F. V. Davenport** (2023). Separating internal and forced contributions to near term SST predictability in the CESM2-LE. *Environmental Research Letters*, 18(10), 104047. <https://doi.org/10.1088/1748-9326/acfdbe>
10. Trok, J. T., **F. V. Davenport**, E. A. Barnes ad N. S. Diffenbaugh (2023). Using machine learning with partial dependence analysis to investigate coupling between soil moisture and near-surface temperature. *Journal of Geophysical Research: Atmospheres*, 128, e2022JD038365. <https://doi.org/10.1029/2022JD038365>
9. Ly, A., **F. V. Davenport** and N. S. Diffenbaugh (2023). Exploring the influence of summer temperature on human mobility during the COVID-19 pandemic in the San Francisco Bay area. *GeoHealth*, 7, e2022GH000772. <https://doi.org/10.1029/2022GH000772>
8. Yu, G., D. B. Wright, and **F. V. Davenport**, (2022). Diverse Physical Processes Drive Upper-Tail Flood Quantiles in the US Mountain West. *Geophysical Research Letters*, 49, e2022GL098855. <https://doi.org/10.1029/2022GL098855>
7. Diffenbaugh, N. S. and **F. V. Davenport** (2021). On the impossibility of extreme event thresholds in the absence of global warming. *Environmental Research Letters*, 16 (11). <https://doi.org/10.1088/1748-9326/ac2f1a>
6. Johnston, E. C., **F. V. Davenport**, L. Wang, J. K. Caers, S. Muthukrishnan, M. Burke, and N. S. Diffenbaugh (2021). Quantifying the effect of precipitation on landslide hazard in urbanized and non-urbanized areas. *Geophysical Research Letters*, 48, e2021GL094038. <https://doi.org/10.1029/2021GL094038>
- ♦ 5. **Davenport, F. V.** and N. S. Diffenbaugh (2021). Using machine learning to analyze physical causes of climate change: A case study of U.S. Midwest extreme precipitation. *Geophysical Research Letters*, 48, e2021GL093787. <https://doi.org/10.1029/2021GL093787>
4. Diffenbaugh, N. S., **F. V. Davenport**, and M. Burke (2021). Historical warming has increased U.S. crop insurance losses. *Environmental Research Letters*, 6(8). <https://doi.org/10.1088/1748-9326/ac1223>
3. Simpson, I. R., K. A. McKinnon, **F. V. Davenport**, M. Tingley, F. Lehner, A. A. Fahad, and D. Chen (2021). Emergent constraints on the large-scale atmospheric circulation and regional hydroclimate: do they still

work in CMIP6 and how much can they actually constrain the future? *Journal of Climate*.
<https://doi.org/10.1175/JCLI-D-21-0055.1>

- ◆ 2. **Davenport, F. V.**, M. Burke, and N. S. Diffenbaugh (2021). Contribution of historical precipitation change to US flood damages. *Proceedings of the National Academy of Sciences*, 118(4).
<https://doi.org/10.1073/pnas.2017524118>
- ◆ 1. **Davenport, F. V.**, J. E. Herrera--Estrada, M. Burke, and N. S. Diffenbaugh (2020). Flood size increases nonlinearly across the western United States in response to lower snow--precipitation ratios. *Water Resources Research*, 56, e2019WR025571. <https://doi.org/10.1029/2019WR025571>

Other Publications

Reports

Technical contributor to Payton, E.A., A.O. Pinson, T. Asefa, L.E. Condon, L.-A.L. Dupigny-Giroux, B.L. Harding, J. Kiang, D.H. Lee, S.A. McAfee, J.M. Pflug, I. Rangwala, H.J. Tanana, and D.B. Wright, 2023: Ch. 4. Water. In: *Fifth National Climate Assessment*. Crimmins, A.R., C.W. Avery, D.R. Easterling, K.E. Kunkel, B.C. Stewart, and T.K. Maycock, Eds. U.S. Global Change Research Program, Washington, DC, USA. <https://doi.org/10.7930/NCA5.2023.CH4>

Articles

"Climate change is making flooding worse: 3 reasons the world is seeing more record-breaking deluges and flash floods", *The Conversation*, 2022

Grants

External Grants Submitted/Pending

Understanding Atmospheric Drivers and Predictability of Flood-Producing Precipitation Events in the United States. PI: Frances Davenport
NASA, Award Amount: \$297,544

Externally Funded as PI

07/2025 – 06/2028 USDA, Agricultural Experiment Station
Using data-driven models to improve forecasts of growing-season water supply in Colorado under a changing climate. PI: Frances Davenport
Award Amount: \$120,000

Internally Funded

02/2026 – 06/2026 Colorado Water Center Enhanced Impacts Initiative
Developing an Interactive Flood Frequency Analysis Dashboard for Colorado Watersheds. PI: Frances Davenport
Award Amount: \$13,252

09/2025 – 12/2025 OVPR Catalyst Set Program
The Centroid: Forging the future of interdisciplinary computational research at CSU
PI: Caitlin Mothes, Co-PIs: Bill Carpenter, Simon Tavener, Frances Davenport, Nikhil Krishnaswamy, and Matthew Ross
Award Amount: \$5,000

07/2025 – 06/2026 School of Global and Environmental Sustainability Curriculum Innovation Grant

Integrating climate change, sustainability and resilience into civil and environmental engineering education. PIs: Pinar Omur-Ozbek and Frances Davenport
Award Amount: \$10,000

11/2024 Borland Equipment Fund
Advancing hydrologic modeling capabilities through GPU-enabled deep learning
Award Amount: \$15,000

09/2022 – 06/2025 Scott Foundation High-Impact Research Fund
Remotely sensed global monitoring of human development amid climate hazards with a multi-year index of human footprint. PI: Patrick Keys, Co-PIs: Frances Davenport, Elizabeth Barnes, Jim Hurrell
Award Amount: \$179,338

Presentations

Selected Workshops and Conference Presentations

- 2025 **(invited talk)** AGU Fall Meeting 2025, “Hydrometeorological Characteristics of High-impact Flood Disasters”, December 15-19, 2025, New Orleans, LA
- 2025 **(invited talk)** Earth and Environmental Science in Action for a Rapidly Changing World Symposium, “Climate Change Attribution: from Long-Term Trends to Individual Events”, Northwestern University, June 6, 2025, Evanston, IL
- 2025 **(invited talk)** National Academies of Science, Engineering and Medicine Extreme Event and Impact Attribution Workshop, “Detection and Attribution of Extreme Precipitation and Flood Impacts”, April 1-2, 2025, Irvine, CA
- 2024 **(invited talk)** AGU Fall Meeting 2024, “Understanding recent trends in regional precipitation extremes: insights from multiple observational datasets and climate models”, Washington, DC
- 2024 **(invited talk)** Workshop on Future Terrestrial Water Availability: Towards an Integrated Perspective on Water, Plants, and Climate, “Extreme Precipitation and Flooding”, Aspen Global Change Institute, Aspen, CO
- 2024 Earth System Predictability Across Timescales Workshop, “Combining Neural Networks and CMIP6 Simulations to Learn Windows of Opportunity for Skillful Prediction of Multiyear Sea Surface Temperature Variability”, National Center for Atmospheric Research, Boulder, CO (poster)
- 2024 **(invited talk)** Chishiki-AI Research Summit, “AI for Climate-Related Hazards and Impacts”, University of Texas at Austin,
- 2024 Workshop on Confronting Earth System Model Trends with Observations: The Good, the Bad and the Ugly, “Assessing historical patterns of regional extreme precipitation change in observations and climate models using quantile regression and machine learning”, National Center for Atmospheric Research, Boulder, CO (poster)
- 2023 **(invited talk)** AGU Fall Meeting 2023, “Attributing damages from extreme climate events under historical and future climate pathways”, San Francisco, CA
- 2023 AMS Annual Meeting 2023, “Transfer Learning for Multi-year Climate Prediction”, Denver, CO (talk)
- 2022 AGU Fall Meeting 2022, “Combining Climate Models, Observations, and Deep Learning for Multi-year Climate Prediction”, Chicago, IL (talk)
- 2022 **(invited talk)** IMSI Economic Impacts of Climate Change Workshop, “Understanding the Costs of Flooding in a Changing Climate”, Chicago, IL
- 2021 AGU Fall Meeting 2021, “Using Machine Learning to Understand Changes in Extreme Precipitation”, New Orleans, LA, (talk)
- 2020 AGU Fall Meeting 2020, “Understanding the Impacts of Climate Change on “Billion-dollar Disaster” Precipitation Events in the United States”, virtual, (talk)
- 2019 AGU Fall Meeting 2019, “Estimating the contribution of historical precipitation changes to increasing flooding damages in the United States”, San Francisco, CA, (talk)

- 2019 Lawrence Berkeley Workshop on Risk Analysis for Extremes in the Earth System, “Historical patterns of precipitation and flood damage in the U.S.”, Berkeley, CA, (poster)
- 2018 AGU Fall Meeting 2018, “Response of flood magnitude to changes in snow precipitation across the western United States”, Washington, D.C., (poster)

Invited Seminars and Presentations

- 2025 Global Energy and Water Exchanges (GEWEX) Flood Crosscutting Initiative Meeting (*virtual*)
- 2025 Dept of Statistics, Colorado State University (Fort Collins, CO)
- 2025 Geophysical Fluid Dynamics Laboratory, National Oceanic and Atmospheric Administration (*virtual*)
- 2024 Physical Sciences Laboratory, National Oceanic and Atmospheric Administration (Boulder, CO)
- 2024 Dept of Civil and Environmental Engineering, University of Colorado Boulder (Boulder, CO)
- 2024 Dept of Geosciences, Colorado State University (Fort Collins, CO)
- 2023 Dept of Atmospheric Science, Colorado State University (Fort Collins, CO)
- 2022 Dept of Atmospheric Science, University of North Dakota (*virtual*)
- 2022 Climate and Global Dynamics Seminar, National Center for Atmospheric Research (Boulder, CO)
- 2022 Dept of Atmospheric and Oceanic Sciences, UCLA (Los Angeles, CA)
- 2022 Dept of Atmospheric Science, University of Illinois, Urbana-Champaign (Champaign, IL)
- 2022 Dept of Atmospheric Science, University of Utah (Salt Lake City, UT)
- 2022 Dept of Civil and Environmental Engineering, Colorado State University (Fort Collins, CO)
- 2022 Dept of Civil and Environmental Engineering, Princeton University (*virtual*)
- 2022 Environmental Science and Engineering, California Institute of Technology (Pasadena, CA)
- 2022 Dept of Geography, University of Georgia (Athens, GA)
- 2022 Dept of Atmospheric, Oceanic, and Earth Sciences, George Mason University (Fairfax, VA)
- 2022 Center for Atmosphere Ocean Science, Courant Institute, NYU (*virtual*)
- 2022 School of Foreign Service, Georgetown University (*virtual*)
- 2021 Dept of Earth System Science, Stanford University (*virtual*)
- 2020 Yosemite Forum (Yosemite, CA, *virtual*)

Teaching

Colorado State University Courses:

**denotes an original course developed by Davenport for CSU students*

CIVE 520: Physical Hydrology

In-person: Spring 2024, Fall 2025, Fall 2026

Offered online: Fall 2025, Fall 2026

Graduate level course covering the fundamental physical processes that shape the movement and distribution of water within the earth system from local to regional to global scales.

*CIVE 480A6: Climate Change Risks and Impacts in Civil and Environmental Engineering

In-person: Fall 2024, Fall 2026

Upper-level undergraduate course on the impacts of climate change for civil and environmental engineering systems. Topics include the scientific basis of climate change, changes to extreme weather and water resources, and incorporating climate change into engineering design. Students are also introduced to using Python to analyze climate and natural disaster data.

Advising and Mentoring

Current CSU Graduate Students

Michael Talbot, PhD student (January 2024 – Present)

Bill Doan, PhD student (August 2024 – Present)

Alexandria Rodgers, PhD Student (August 2024 – Present), *co-advised with Antonio Meira Neto*

Previous CSU Graduate Students

Nicole Keeney, MS student (graduated Fall 2025)

Postdoctoral Mentees

Bryam Orihuela Pinto (July 2023 – July 2025), *co-advised with Pat Keys (Atmospheric Science)*

Undergraduate Research Mentees

Brigid Neuheardt (Spring 2025), CSU Scott Undergraduate Research Experience

Rob Iliff (Spring 2025), CSU Scott Undergraduate Research Experience

Madisyn Bietz (Spring 2023), CSU Scott Undergraduate Research Experience, *co-advised with Pat Keys*

Honors and Awards

2026 Outstanding Junior Faculty Performance Award, CSU Dept. of Civil & Environmental Engineering

2025 Borland Chair of Hydrology, CSU Dept. of Civil & Environmental Engineering

2021 Graduate Student Award for Research or Scholarly Achievement, Stanford Earth System Science Dept.

2020 Neukermans Family Stanford Interdisciplinary Graduate Fellowship (3 years graduate funding)

2020 ARCS Foundation Scholar Fellowship (declined)

Service and Outreach

Department Service

Member, Graduate Instruction Committee, Civil and Environmental Engineering (Fall '25, Fall '26)

Member, Borland Committee, Civil and Environmental Engineering (Fall '25 – Present)

Faculty Advisor, Water Science and Engineering Seminar Series (Spring '25 – Fall '25)

Member, Faculty Search Committee, Civil and Environmental Engineering (Fall '24 – Spring '25)

University Service

Member, Centroid Research Council (Fall '26 – Present)

Professional Service

Associate Editor, *Journal of Climate*

Grant Referee for *National Oceanic and Atmospheric Administration (Climate Variability & Predictability Program; Modeling, Analysis, Predictions, & Projections Program)*, *Climate Change AI Innovation Grants*, *National Science Foundation Hydrologic Sciences Program*

Journal Referee for *PNAS*, *Science Advances*, *Geophysical Research Letters*, *Water Resources Research*, *Environmental Research Letters*, *Nature Communications*, *JGR: Atmospheres*, *Artificial Intelligence for the Earth Systems*, *Hydrology and Earth System Sciences*, *Earth's Future*, *Journal of Hydrology*, *Climatic Change*, *Hydrological Processes*, *Natural Hazards*

Media Engagement

Engagement with local and national media outlets on published research and issues related to climate change and extreme events. Coverage includes: *Washington Post*, *San Francisco Chronicle*, *Bloomberg*, *Reuters*, *ABC News*, *CBS News*, *CNBC*, *National Geographic*, *Scientific American*, *Yale Climate Connections*

Selected coverage:

Washington Post, “*Before the floods, Asheville was called a ‘climate haven.’ Is anywhere safe?*”, 10/01/24

National Geographic, “*Drought, flooding, drought again: Is ‘weather whiplash’ our new normal?*”, 9/18/24

New York Times, “*Climate change is making floods and landslides more likely, study finds*”, 6/28/23

SF Chronicle, “*Floods have devastated parts of California. Here’s why they’re so hard to forecast*”, 3/24/23

Boise State Public Radio, "*How climate change is feeding big floods around the Mountain West*", 7/19/22
Desert Research Institute, "*Study Explores Uncertainties in Flood Risk Estimates*", 6/14/22
Stanford News, "*Stanford researchers use artificial intelligence to unlock extreme weather mysteries*", 8/10/21
ABC News, "*1-in-100-year floods happening so often, the term may change*", 8/5/21
CBS News, "*Climate change is responsible for billions of dollars in flood costs, study says*", 1/11/21
Stanford News, "*More rain and less snow means increased flood risk, Stanford study reveals*", 1/27/2020

Professional Affiliations

member, American Geophysical Union (2018 - Present)
member, American Meteorological Society (2022 - Present)

Last Updated October 7, 2026